

SIMATS ENGINEERING

Assessment Tool 2: Scenario-Based Research Assessment

Course: Artificial Intelligence and Expert Systems (MLA01)

The Waymo Driver: Fully Autonomous Ride-Hailing at Commercial Scale

Domain: Autonomous Vehicles

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1. Title of AI Application

For this assignment I chose to study the Waymo Driver, the self-driving system built by Google's sister company Waymo that now powers a real, paid robotaxi service in several U.S. cities. It is one of the few AI systems in this space that has actually gone from research prototype to something ordinary people use every day, which is what made it an interesting pick.

2. Domain Selection

Domain: Autonomous Vehicles. More specifically, this report looks at driverless ride-hailing, where a car picks up and drops off a passenger with nobody sitting behind the wheel.

3. Introduction and Background

Waymo did not appear overnight. It started life in 2009 as Google's self-driving car project, spent years quietly testing on closed courses and public roads, and was eventually spun off as an independent Alphabet company in 2016. Today it runs what is probably the most widely used driverless ride-hailing service anywhere in the world. A rider opens the Waymo app (or, in a few cities, hails one through Uber), requests a trip, and a car with nobody in the driver's seat shows up and takes them where they need to go.

What sets Waymo apart from the driver-assist features found in many modern cars is the level of autonomy involved. This is SAE Level 4: inside its mapped, geofenced service areas, the car handles the entire driving task on its own – sensing, deciding, and steering – with no human ready to take over. By the middle of 2026 the company had completed more than 20 million paid trips and over 220 million rider-only miles, and it was running a fleet of somewhere between 3,000 and 4,000 vehicles across more than ten American metro areas, with early pilots also getting underway in London and Tokyo.

4. Problem Identification

Most road accidents come down to human error – people driving tired, distracted, or impaired. That is a huge and fairly stubborn problem. On top of that, conventional taxi and ride-hailing businesses are limited by how many human drivers they can find and afford, which pushes up prices and creates service gaps whenever drivers are in short supply. City transport systems also lose out because human driving is inconsistent and gives planners very little real-time data to work with.

So the problem Waymo is really trying to solve is this: can a vehicle be engineered to drive more safely and predictably than the average human, and can that vehicle then be deployed commercially, at the scale of an entire city, without needing a driver at all?

5. Need for AI Solution

You cannot solve this with a simple set of hard-coded rules. Real streets are far too unpredictable – a pedestrian steps out unexpectedly, a construction crew redirects traffic, an ambulance needs to get through, the weather turns bad – and no fixed rulebook can cover every one of those situations in advance. This is exactly where AI, and deep learning in particular, earns its place: instead of being told what to do for every possible scenario, the system learns patterns from millions of miles of driving data and applies that learning to situations it has never technically seen before.

An AI-driven car also does not get tired, distracted, or impatient. It can react within milliseconds by fusing information from multiple sensors, and it keeps improving as more driving data flows back into training. In short, AI is not just a nice-to-have here; it is more or less the only realistic way to get driverless operation working reliably at city scale.

6. AI Technologies Used

A number of different AI techniques work together inside the Waymo Driver's software stack:

- Computer vision: deep convolutional and transformer-based neural networks process the camera feeds to spot and classify pedestrians, vehicles, cyclists, traffic lights, and road signs.
- Sensor fusion: separate machine learning models pull together lidar point clouds, radar readings, and camera images into one combined picture of what is happening around the car.
- Behaviour prediction: deep learning models try to anticipate what pedestrians, cyclists, and nearby vehicles are likely to do next, not just where they are right now.
- Planning and decision-making: learned models are combined with search and optimisation techniques to work out a safe, efficient path, and this plan is recalculated continuously as the scene changes.
- LLMs and vision-language models: Waymo has recently started folding in large language model and vision-language model reasoning, giving the car a bit more “common sense” when it runs into an unusual or ambiguous situation.

- Simulation and reinforcement learning: on top of real-world driving, models are trained and stress-tested across billions of simulated miles, with a feedback loop that resembles reinforcement learning.

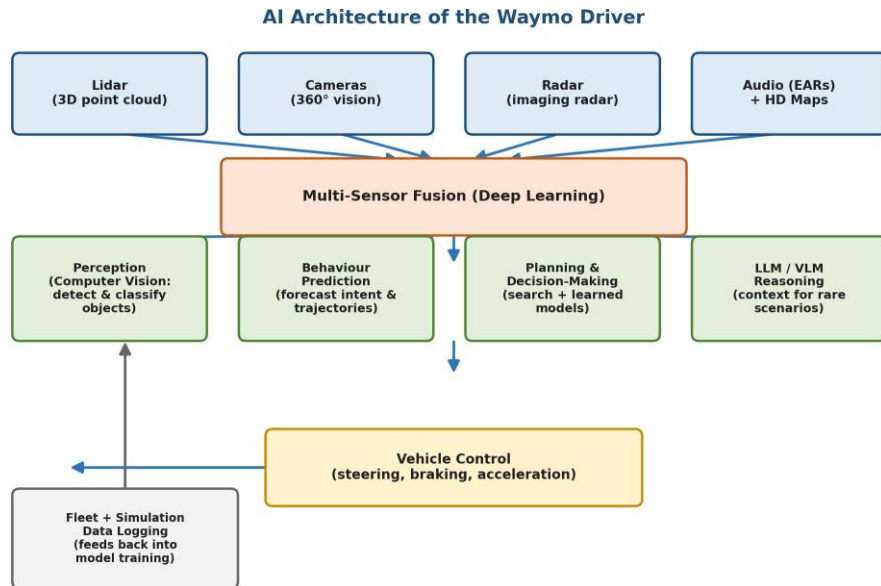


Figure 1: How the pieces fit together – sensor inputs, fusion, the perception/prediction/planning layer, control output, and the feedback loop back into training.

7. Working Mechanism

Waymo has a nice shorthand for how the whole thing works: “Sense, Solve, Go.”

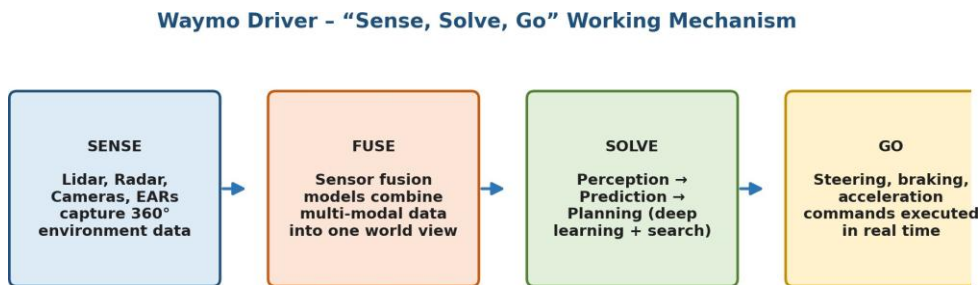


Figure 2: The Sense → Fuse → Solve → Go loop, which repeats many times a second while the car is driving.

- Sense: a sensor pod on the roof combines lidar, high-resolution cameras, imaging radar, and external microphones (Waymo calls these EARs) to build a full, long-range, 360-degree picture of the surroundings, in daylight or darkness and across a range of weather conditions.
- Solve: an onboard computer fuses all that sensor data together and runs deep learning models in real time to figure out what is around the car, what it is about to do, and what the car itself should do about it. The different sensor types are also used to double-check

each other – lidar, for example, can confirm whether something a camera flagged as a stop sign is real or just a reflection in a shop window.

- Go: once a path is chosen, the system sends out the actual steering, acceleration, and braking commands, and it keeps re-checking the scene many times per second so it can react almost instantly if, say, a pedestrian steps into the road or a siren starts up nearby.

Interestingly, the car does not need a live cell signal to make any of these driving decisions – all the sensing and reasoning happens onboard. The network connection is mainly there for things like fleet dispatch, remote assistance when the car hits an edge case it wants a second opinion on, and pushing out updated maps.

Sensor Architecture: How the Sensors Work

It's worth zooming in on the sensor side specifically, since this is really the foundation everything else in the stack is built on. The Waymo Driver does not rely on any single sensor type – it deliberately overlaps several different kinds of sensing so that the weaknesses of one are covered by the strengths of another.



Figure 8: Top-down schematic of the sensor suite – lidar (360° ring), camera fields of view, front/rear radar wedges, and audio/corner sensors, all feeding into a single fused world model.

- Lidar spins continuously to send out laser pulses in every direction and measures how long they take to bounce back, building a precise 3D point cloud of everything nearby, out to roughly 300 metres, regardless of whether it's day or night.
- Cameras provide full 360-degree colour vision with high dynamic range, which is what lets the system actually read stop signs, traffic-light colours, lane markings, and text – things lidar and radar cannot interpret on their own.
- Radar fills in the gap both lidar and cameras struggle with: it can instantly measure how fast something is moving and in what direction, and it keeps working reasonably well in rain, fog, or snow that would otherwise degrade a camera or lidar signal.
- External audio receivers (EARs) and short-range corner sensors pick up on things a purely visual system would miss entirely, like an approaching siren from around a blind corner.

None of these run in isolation, though. Their outputs are combined continuously by the fusion layer, so if a camera flags something ambiguous – say, what might be a stop sign reflected in a shop window – the lidar and radar readings for that same spot can confirm or rule it out almost instantly. Below are two close-up views of the actual sensor hardware, cropped from the vehicle photos used earlier in this report.



Figure 9: Close-up of the roof-mounted sensor housing, containing the primary long-range lidar unit along with several cameras.



Figure 10: Close-up of a front fender-mounted sensor pod, one of several smaller pods placed around the body for close-range coverage and reduced blind spots.

Earlier Mechanical Lidar Designs

Before looking at the solid-state design Waymo uses now, it helps to see what came before it. Most early automotive and robotics lidar units were mechanical: a laser and receiver sit inside a housing that itself spins, or a small internal mirror tilts and rotates to steer the beam, sweeping it across the scene one line at a time.

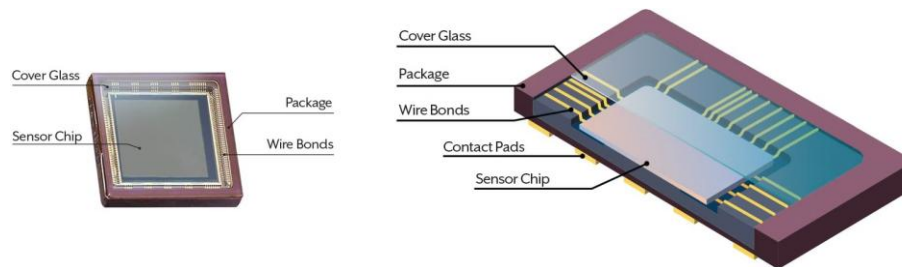


Figure 11: A commercial mechanical lidar unit (left) and its internal layout (right) -- a motor with an angle encoder turns a rotating mirror, which steers an outgoing laser beam and directs the reflected echo back onto a photodiode receiver.

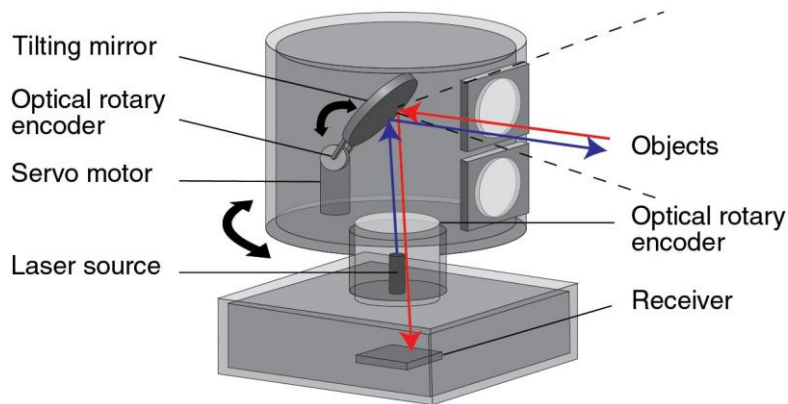


Figure 12: A tilting-mirror lidar design, where a servo motor tilts an angled mirror while the whole head rotates, and an optical rotary encoder tracks the mirror's exact angle so each returning pulse can be mapped to a precise direction.

Designs like these work well and were the industry standard for years, but the motor, mirror, and encoder are moving parts that add cost, size, and long-term wear -- which is exactly the gap the solid-state design below is built to close.

Solid-State Lidar: How It Works and Why It Was Chosen

The lidar unit described above no longer relies on the older spinning-mirror design (the kind shown in earlier research lidars, where a servo motor physically rotates a mirror and the whole housing to sweep a laser beam across the scene). Waymo's current sensor hardware instead uses solid-state lidar, where the entire unit is built from a fixed stack of electronic layers with nothing mechanical spinning at all.

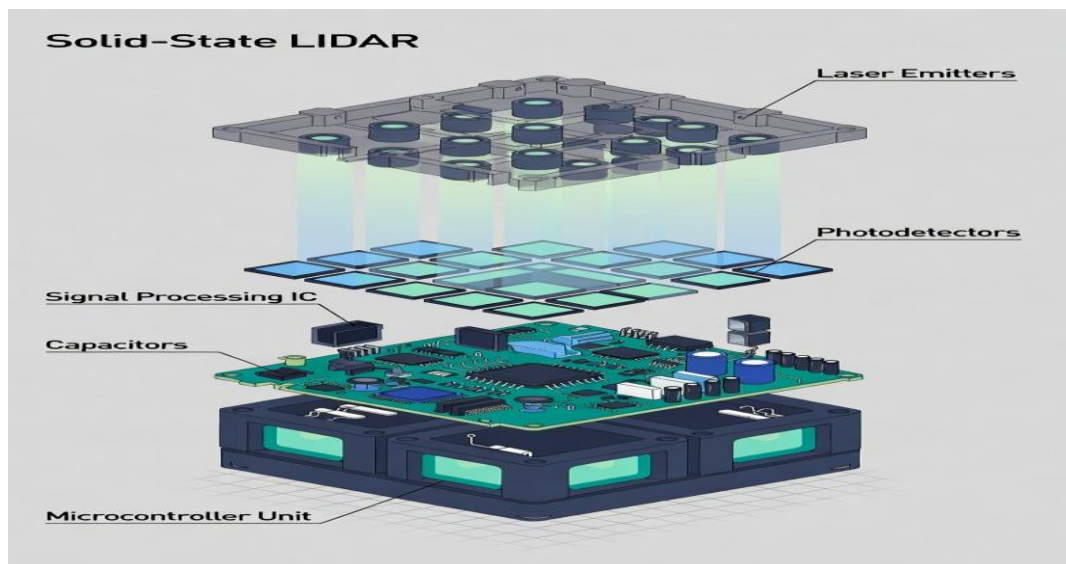


Figure 13: Exploded view of a solid-state lidar unit -- a fixed array of laser emitters and photodetectors stacked above a signal-processing board, with no rotating parts.

How it works:

- Instead of one laser and one spinning mirror, a fixed grid of laser emitters fires in sequence, each one covering a slightly different angle, so the whole field of view is illuminated without any part of the housing physically turning.
- A matching array of photodetectors sits underneath the emitters and reads back how long the reflected light takes to return, building the same kind of distance measurements and 3D point cloud that a mechanical lidar produces, just through fixed optics rather than a rotating head.
- A signal-processing IC on the board timestamps each returning pulse and converts it into a range value, while nearby capacitors supply the short, high-current bursts the laser emitters need each time they fire.
- A microcontroller unit ties the whole board together, sequencing which emitters fire, reading back the matching detectors, and packaging the resulting point cloud before handing it to the vehicle's main sensor-fusion computer.

Why this was chosen over other lidar designs:

- No moving parts: older lidar designs relied on a servo motor turning a mirror or the whole housing to sweep the beam around; removing that mechanical assembly means there is nothing to wear out, vibrate loose, or fall out of alignment after years of driving.
- Cheaper to build at scale: a fixed emitter/detector array can be produced with standard semiconductor and PCB assembly lines rather than precision-machined rotating optics, which is part of why newer, sixth-generation sensor hardware costs less per vehicle than earlier spinning units.
- Smaller and more rugged: without a rotating core, the assembly can be flatter and more easily sealed against dust and moisture, which suits the many small fender-mounted and corner sensor pods Waymo uses rather than only a single large roof-mounted unit.
- Faster, targeted refresh: because each emitter/detector pair is addressed electronically rather than waiting for a mechanical sweep to come back around, the system can re-check specific zones, such as directly ahead of the car, more often, which helps it react quickly if a pedestrian steps into the road.

Radar Hardware

The radar units mentioned earlier are built the same way as the lidar unit, layer by layer: a radome (the outer cover) protects an internal absorber and antenna PCB, which sits on heat-dissipation pads above the main housing, with the connectors that link back to the vehicle's wiring at the base.

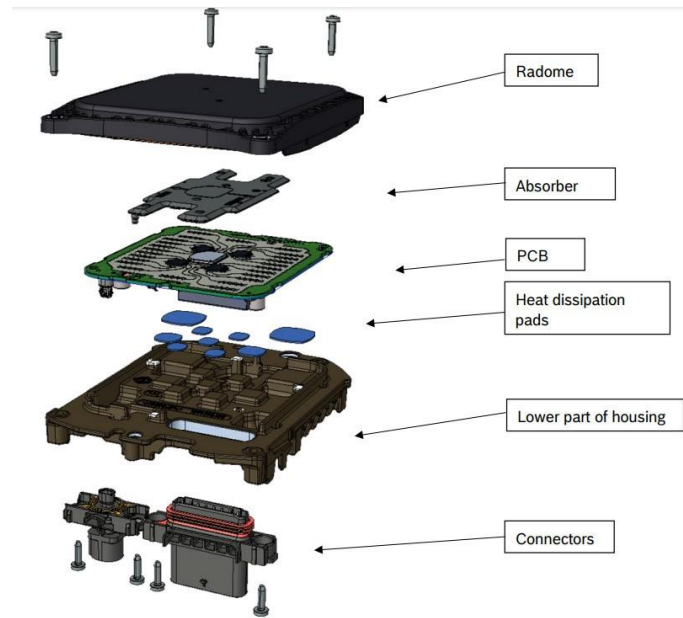


Figure 14: Exploded view of an automotive radar unit -- radome, absorber, antenna PCB, heat-dissipation pads, lower housing, and connectors, stacked into a single sealed module.

Radar hardware has also gone through generational changes of its own. Newer long-range radar units move the printed circuit board and its separate connector into a die-cast body with its own backplane, rather than the simpler plastic backplane-and-connector arrangement used in second-generation units, giving better range and resolution in the same footprint.



Figure 15: Second-generation versus third-generation long-range automotive radar -- the newer unit (right) integrates the PCB and connector into a die-cast body for better performance in a similar package size.

Camera / Image Sensor (CMOS)

The cameras in the sensor suite rely on CMOS image sensors. Light passes through a transparent die and a transparent conductive oxide (TCO) layer before reaching a light-sensitive semiconductor layer, which converts photons into an electrical charge at each pixel. Metallic

interconnects then carry that charge down to a silicon die containing the CMOS circuits, which reads out each pixel and assembles the full image frame -- all of this bonded together in a single flip-chip package, which is what lets the sensor stay compact enough to fit inside the small camera housings mounted around the vehicle.

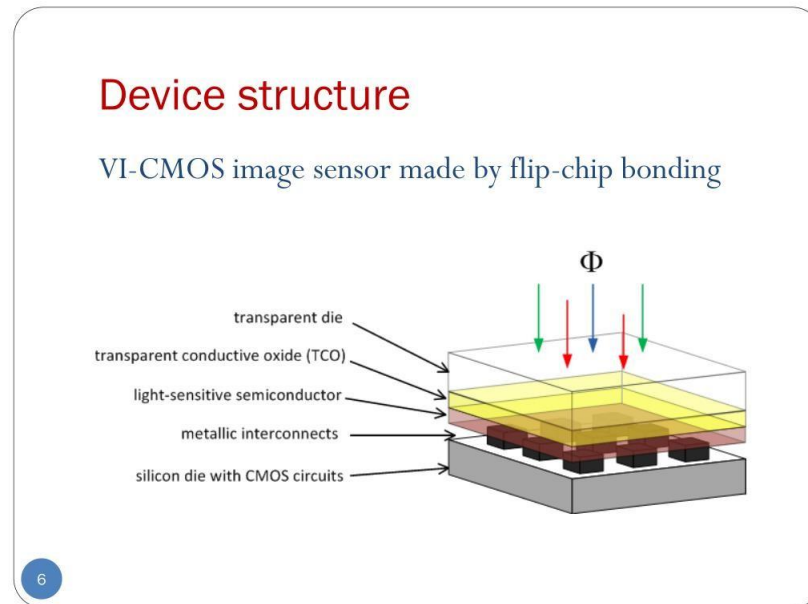


Figure 16: Cross-section of a CMOS image sensor made by flip-chip bonding, showing the transparent die, TCO layer, light-sensitive semiconductor, metallic interconnects, and the underlying silicon die with CMOS readout circuits.

8. Data Sources and Processing

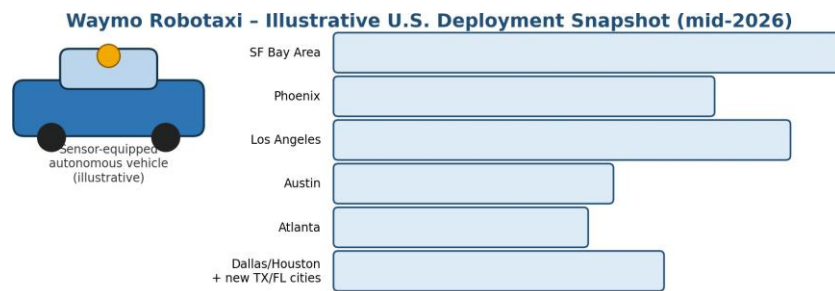
Two broad kinds of data feed Waymo's models. The first is real driving data collected from the fleet over millions of on-road miles, which captures all the messy, genuine sensor noise and rare situations that happen out in the world. The second is simulated data from Waymo's own internal simulation environment, which by now has clocked up tens of billions of virtual miles. Simulation is particularly useful for the dangerous, rare edge cases – a child suddenly running into the street, for instance – that would be far too risky to try to collect for real.

Raw sensor data such as lidar point clouds, camera frames, and radar returns gets labelled and fed into an automated machine-learning pipeline that searches for and validates better model architectures before anything is pushed out to the fleet. Alongside this, Waymo relies heavily on pre-built, high-definition maps of every city it operates in, which give the car centimetre-level knowledge of lane markings, signage, and intersections well before it ever drives through them.

9. Real-World Implementation

This is not a lab demo – Waymo runs a genuine, paid, commercial service. By mid-2026 it was offering fully driverless rides in San Francisco, Los Angeles, Phoenix, Austin, Atlanta, Miami, Dallas, Houston, San Antonio, Orlando, and Las Vegas, with more U.S. cities and its first international markets (Denver, San Diego, Tampa, London, and Tokyo among them) at various

stages of testing or rollout. Riders can book through the Waymo One app directly, or in some markets through Uber.



Bars show relative service presence across active U.S. markets (illustrative, not to scale)

Figure 3: Illustrative snapshot of Waymo's autonomous vehicle and its relative service presence across active U.S. markets (not an actual photograph; drawn for illustration only).

The company reports having completed more than 20 million paid autonomous trips and over 220 million rider-only miles, with weekly paid trips somewhere in the 400,000–500,000 range and a stated goal of hitting around a million rides a week by the end of 2026. A large new funding round has also gone toward scaling up manufacturing of its sixth-generation vehicle platform, with plans to eventually build tens of thousands of vehicles a year.

A few real photographs help make this more concrete – what the vehicle and the service actually look like on the street:

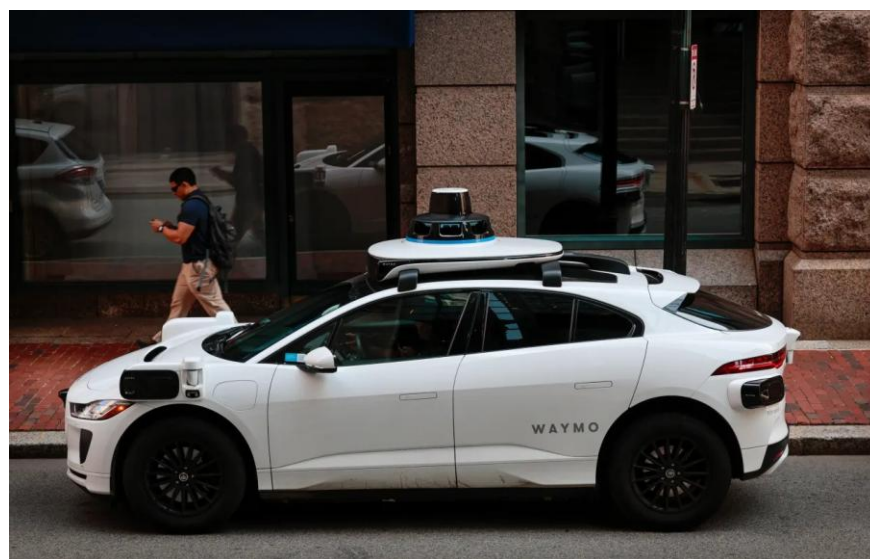


Figure 4: Rear view of a Waymo vehicle showing its roof-mounted sensor housing – lidar, radar, and cameras – as it drives through a city street.



Figure 5: A Waymo vehicle (built on the Jaguar I-PACE platform) parked curbside while a pedestrian walks past, showing the compact sensor pods mounted on the front fender and roof.



Figure 6: A passenger stepping out of a next-generation Waymo robotaxi at a curbside pickup and drop-off point.



Figure 7: A Waymo autonomous vehicle in everyday operation on public roads.

10. Impact and Benefits

A few benefits stand out from the data Waymo and outside observers have published so far:

- **Safety:** Waymo publishes its own safety data, which it says shows lower rates of certain crash types compared with human-driven benchmarks over its accumulated rider-only miles.
- **Accessibility:** because the service does not depend on a human driver being available, it can run around the clock and may open up mobility options for people who cannot or do not drive themselves.
- **Scale without a labour bottleneck:** the fleet can, in principle, be expanded independently of how many human drivers are willing to work, which is not true of traditional ride-hailing.
- **Useful city-level data:** anonymised fleet telemetry can give city planners a better sense of real traffic and mobility patterns.
- **Longer-term economic effects:** some analysts think shared autonomous fleets could eventually reduce how many cars households in dense cities feel they need to own.

11. Challenges and Limitations

It would be misleading to only tell the success story, though – there are real challenges here too:

- **Cost:** Waymo's sensor suite, with multiple lidars, radar units, and cameras, is considerably more expensive than the camera-only approach some competitors use.
- **Operational hiccups:** vehicles have occasionally gotten stuck (Waymo's own term is “bricked”) during traffic-signal outages or severe weather, causing localised traffic jams, and there have been isolated but serious incidents, including a reported collision near a school that drew regulatory attention.

- Regulatory scrutiny: U.S. safety regulators such as the NHTSA and NTSB have opened investigations into specific incidents, including reports of vehicles not stopping for school buses.
- Trust and transparency: lawmakers have pushed Waymo on exactly how much human “remote assistance” actually happens behind the scenes, and surveys still show a large chunk of the public feels nervous about riding in a car with no driver.
- Geofencing limits: the system only works within pre-mapped service areas rather than anywhere a person could legally drive, and expanding into a new city takes a long stretch of mapping and validation first.
- Ethical and workforce questions: as this scales up, it naturally raises questions about what happens to professional driving jobs, and about how liability should be assigned if something does go wrong.

12. Future Scope and Emerging Trends

Looking ahead, a few directions seem likely to shape where this technology goes next:

- Bringing in more LLM and vision-language model reasoning so the car can handle unusual, one-off scenes with something closer to human-style judgement.
- Pushing into new environments – highway driving, extreme winter weather, and international cities such as London and Tokyo.
- Cheaper, more streamlined sensor hardware (the sixth-generation Waymo Driver is already a step in this direction) to bring per-vehicle costs down without giving up on safety.
- Closer integration with existing ride-hailing and delivery platforms, plus knock-on effects for how cities think about parking and road infrastructure as fleets grow.
- More public safety data and independent audits, which will likely be necessary to keep building trust as the service expands into more places.

13. Conclusion

Working through this case study, what stands out most to me is how many different AI techniques have to work together just to make a single safe left turn – computer vision, sensor fusion, behaviour prediction, and planning, all trained on a mix of real and simulated driving data. Waymo's rapid expansion through 2025 and 2026 shows that Level 4 autonomy has genuinely moved past the demo stage into a scaling commercial service. At the same time, the incidents, regulatory investigations, and lingering public unease are a reminder that safety validation, transparency, and cost are still very much open problems, not solved ones. If anything, Waymo's story is a good example of how a real-world AI system can be impressive and still very much a work in progress at the same time.

Related Academic Literature

Company blog posts only tell part of the story, so it is worth looking at what the academic literature says about the underlying techniques. A handful of recurring themes came up across the papers I looked at:

- Deep learning for perception: Wen and Jo (2022) reviewed deep-learning-based perception pipelines for autonomous driving – covering 3D object detection and traffic-sign/light recognition – and pointed out that detecting partially occluded road users is still genuinely hard, which lines up with why Waymo leans so heavily on combining cameras, lidar, and radar rather than trusting any single sensor.
- Sensor and data fusion: Zhu et al. (2023, on arXiv) surveyed deep-learning approaches to multi-sensor data integration in self-driving perception and proposed a taxonomy built around multi-view, multi-modality, and multi-frame fusion – essentially the same class of problem Waymo is solving when it cross-checks lidar, radar, and camera streams against one another.
- Deep learning across the whole driving stack: a couple of IEEE surveys from 2019 and 2021 look at how deep learning has been applied to vehicle control, path planning, computer vision, and sensor fusion, which helps explain why deep learning has largely replaced older, hand-coded rule systems in this field.
- Motion planning and reinforcement learning: a 2020 arXiv survey reviews deep reinforcement learning approaches to motion planning, including car-following, lane-keeping, and driving in dense traffic – all relevant to how planning modules like Waymo's get trained and evaluated in simulation before they ever touch a real road.
- Trajectory and behaviour prediction: a further IEEE survey focuses specifically on deep-learning-based vehicle trajectory prediction, reviewing learning-based forecasting methods for other road users – the same job Waymo's behaviour-prediction models are doing.
- Broader surveys tying it all together: one wider survey on deep and reinforcement learning in autonomous driving covers scene understanding, motion planning, decision-making, and control as a single pipeline, which maps almost directly onto what Waymo calls “Sense, Solve, Go.”

Put together, this literature suggests that none of the individual pieces inside the Waymo Driver are exotic or unique to one company – deep-learning perception, sensor fusion, behaviour prediction, and learned planning are all active, well-studied research areas. What Waymo has really done is take techniques that researchers have been refining for years and turn them into a system reliable enough to run commercially, at scale, on public roads.

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